Assignment 1. Random Numbers Marks 25

Posted on 12.08.2026 @ 2:30 pm and due on 13.08.2026 @ 5:40 pm

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1. Use the iterative equation $x_{i+1} = c\,x_i\,(1 - x_i)$ to generate 1,000 random numbers. Show the correlation among them by plotting x_i vs x_{i+k}. Use seed $x_0 = 0.1$ but choose your own three (3) different c. Try various k, say 3, 5 and 10. **[5]**

2. Write your own LCG random generator with the following set of parameters $a = 1103515245$, $c = 12345$, $m = 32768$ and again check for correlation by plotting for $k = 5$. `You must store your LCG code in the library file for all future use till Endsem.` **[5]**

3. Determine the value of π using *throwing method* by choosing a quarter circle of unit radius in the first quadrant. Plot the value of π versus number of throws $20 \leq N \leq 5,000$. **[5]**

4. Simulate the radioactive decay $A \to B \to C$ for the decay constants $\lambda_A = 0.0347$ and $\lambda_B = 0.0198$. Initial number of nuclei $N_A = 500$, $N_B = 0$, $N_C = 0$. **[5]**

5. Generate pRNG having exponential distribution of the form $\exp(-x)$ from pRNG having uniform distribution in $[0, 1)$. Generate at least 5,000 random numbers **[5]**